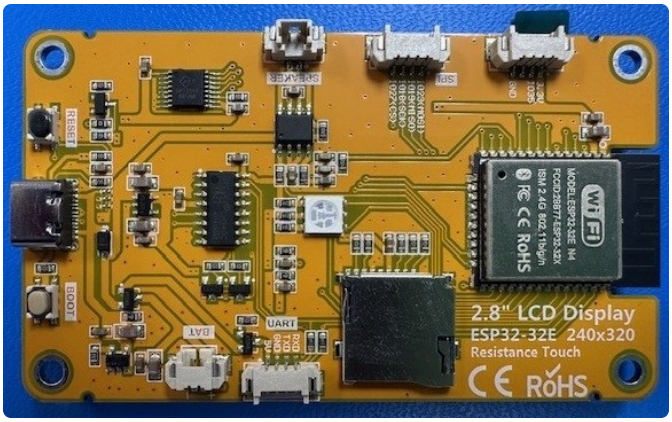
SHACK PROJECTS · EXTENDED EDITION

The Cheap Yellow Display

A $15 shack gadget: what it is, what to buy, and how to turn it into a HamClock-style dashboard

By Ed Barnes · November 3, 2026



The back of the author's Elegoo board: ESP32-WROOM-32E module (right), microSD slot (bottom center), battery and UART connectors (bottom), RESET and BOOT buttons beside the USB-C port (left), and speaker and SPI connectors (top). Photo: author.

AT A GLANCE

Cost	Setup	You need	Power	Best firmware
About $14 per board (Elegoo 2-pack, $27.99)	About 10 minutes with a web flasher, no coding	Chrome or Edge, a USB data cable, your callsign and grid square	Under 1 W from 5 V USB	CYD Ham Dashboard by HenrysCat

Why a $15 board belongs in the shack

For about the price of a coax jumper, you can buy a Wi-Fi computer with a color touchscreen already attached. Hobbyists call it the Cheap Yellow Display, or CYD, after the yellow circuit board on the back.

For hams, that adds up to a small, low-power screen that can sit on the operating desk and show the things we check all day: UTC time, solar and band conditions, DX spots, satellite passes, and station status. No Raspberry Pi, no monitor, no SD card to corrupt.

This extended edition covers what the board is, which version to buy, how to load software without writing code, a survey of ham-friendly projects, and a step-by-step guide to turning one into a HamClock-style dashboard.

What is the CYD?

The CYD is an ESP32 microcontroller, a 2.8-inch 240×320 color touchscreen, and a microSD slot on one board. The name comes from Brian Lough (witnessmenow), whose GitHub repository became the community hub for the original Sunton ESP32-2432S028R. This article uses the Elegoo 2.8-inch ESP32-32E display (model E32R28T), a close cousin with the same screen and touch wiring, tested by the author with the CYD Ham Dashboard.

Feature	Elegoo 2.8-inch ESP32-32E (E32R28T)
Processor	ESP32-WROOM-32E module, dual-core 240 MHz, Wi-Fi and Bluetooth
Flash	4 MB
Display	2.8-inch, 240×320, TN panel, ILI9341 driver over SPI
Touch	Resistive, XPT2046 controller
Storage	microSD slot
USB	Single USB-C; enters download mode automatically
Battery	3.7 V LiPo connector with onboard charger
Extras	RGB LED, speaker connector, SPI and serial connectors
Power	5 V, about 0.65 W with the display on
Board size	50 × 86 mm
Price	$27.99 for a 2-pack on Amazon (Nov 2, 2026)

Source: LCDWIKI E32R28T specification (PDF); price checked by the author. LCDWIKI (Shenzhen QDtech) designs this board; Elegoo sells it under its own name, so you will see both names on spec sheets.

Two limits matter for shack projects. First, few GPIO pins are free; the SPI and serial connectors are the easiest way to add sensors. Second, the TN panel looks best head-on, so mount it facing the operating position.

Want a bigger screen? You may want to order the 4-inch version instead, for about $5 more per board. Its 480×320 display shows larger text and more DX and POTA spots, and the CYD Ham Dashboard flasher has a 4-inch (ST7796S) build. The author has not tested the 4-inch board, so check that the listing names the ST7796S driver.

How it compares with the original CYD

The Elegoo board and the original Sunton ESP32-2432S028R use the same screen, the same touch controller, and the same pins for both, so CYD firmware built for the ILI9341 panel runs on either. The differences are in the extras.

Feature	Classic CYD (ESP32-2432S028R)	ESP32-32E (E32R28T)
Module	ESP32-WROOM-32	ESP32-WROOM-32E, 4 MB flash
Display	2.8" ILI9341, SPI on GPIO 14/13/12, CS 15, DC 2	Same panel and same pins
Backlight	GPIO 21	GPIO 21
Touch	XPT2046 on GPIO 25/32/39, CS 33, IRQ 36	Same controller and same pins
USB	Micro-USB, or micro-USB plus USB-C on later boards	Single USB-C with automatic download mode
Battery	None	3.7 V LiPo connector with onboard charger
Light sensor	LDR on GPIO 34	None; GPIO 34 reads battery voltage
RGB LED (red)	GPIO 4	GPIO 22; GPIO 4 enables the audio amplifier
Extra connectors	Extended IO headers	SPI peripheral, serial, and IO35 input connectors
Power	About 0.6–0.9 W	0.65 W display-only (130 mA at 5 V)

Sources: LCDWIKI E32R28T specification (PDF), atomic14 CYD page, Home Assistant community thread on the E32R28T.

What that means in practice:

- **Use the ILI9341 build** of CYD firmware. The author's E32R28T runs HenrysCat's CYD Ham Dashboard with this build. Newer boards marked E32R28T-1 use an ST7789 panel and need the ST7789 build.
- **Expect small differences.** Features tied to the light sensor or the red LED may not behave the same, because those pins do different jobs on this board.
- **The battery connector is a bonus for field use.** A small 3.7 V LiPo turns the dashboard into a portable POTA or Field Day display.
- **Flashing is simpler.** The USB-C port enters download mode on its own, so you rarely need the BOOT button.
- **Buy by model number.** The author's board is an Elegoo, sold on Amazon as a 2-pack. Search for "Elegoo ESP32 2.8", "E32R28T" or "2.8inch ESP32-32E" to get this board, or "ESP32-2432S028R" for the classic.

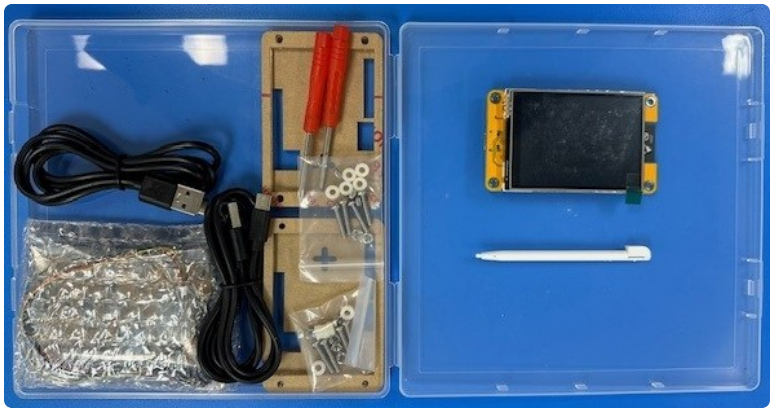
Where to buy and what to pay

Buy the Elegoo ESP32-32E (E32R28T) 2.8-inch display on Amazon: $27.99 for a 2-pack, about $14 each, as of Nov 2, 2026. It is the board the author tested with the CYD Ham Dashboard, and it arrives in 1–3 days with easy returns.

Search Amazon for "Elegoo ESP32 2.8" or "E32R28T" to find it. Price checked by the author; Amazon prices change often.



The Elegoo 2-pack as sold on Amazon. Photo: author.



What's in the box: the display, acrylic case panels with mounting screws and standoffs, two small screwdrivers, USB cables, connector leads, and a touch stylus. Photo: author.

For a bigger screen, you may want to order the 4-inch version for about $5 more per board; see the note under "What is the CYD?" before you buy.

A club group buy works well here: a 2-pack or 5-pack lowers the per-board cost, and having a few identical boards makes a build night much easier to support.

Getting started: four ways in

Most hams should start with a web flasher: no code, no toolchain, about five minutes. The other three paths are there when you want to customize.

Path	Skill needed	What you do	Good for
Web flasher	None	Plug in USB, open a web page in Chrome or Edge, click Install	Ready-made dashboards and tools
Arduino IDE / PlatformIO	Some C++	Install ESP32 board support plus the TFT_eSPI and XPT2046 libraries, then edit and upload sketches	Custom displays, learning
ESPHome	YAML config files	Describe the screen and sensors in a config file; pairs with Home Assistant	Shack monitoring and automation
MicroPython	Python	Flash MicroPython, then write scripts	Quick experiments

A few things trip up first-timers:

- Use a data USB cable, not a charge-only one. If the computer does not see a new serial port, swap the cable first.
- The board uses a USB-to-serial chip (often a CH340). Windows and macOS usually have the driver; if not, install the CH340 driver from the chip maker.
- Web flashers use the Web Serial feature, which works in Chrome and Edge on a desktop but not in Firefox or Safari.
- If flashing stalls at "Connecting...", hold the BOOT button on the board while it starts, then release.

For the Arduino path, Random Nerd Tutorials and the witnessmenow repository have tested example sketches.

Ham radio projects for the CYD

Several open-source projects turn a CYD into a shack appliance with nothing more than a web flasher. The table below lists the ones most useful to a radio club, starting with the closest thing to HamClock.

Project	What it shows	Install	Link
CYD Ham Dashboard (HenrysCat)	Seven pages: UTC/local clock, HF propagation, VHF conditions, greyline map, PSKReporter map, DX cluster spots, POTA spots	Web flasher or PlatformIO	GitHub
cyd-dashboard (Matt Housley)	ISS tracker, PSKReporter, FT8/DX/POTA/SOTA spots, HF propagation, weather and news	PlatformIO	Trendshift listing
9M2PJU ESP32 DX Cluster Client	Live telnet DX cluster spots on many ESP32 screens	Web flasher	hamradio.my
Overhead (JamesDavid)	Satellites overhead, rocket launches, ADS-B aircraft, space weather	Web flasher	Trendshift listing
CYD Weather Dashboard (HenrysCat)	5-day forecast, hourly timeline, sun and moon, alerts	Web flasher	Trendshift listing
ESPHome on the CYD	Any sensor you wire up: shack temperature, power supply voltage, rotator heading, via Home Assistant	YAML config	ESPHome examples

Ideas for club build nights that go beyond ready-made firmware:

- A net control timer and check-in counter, with the touchscreen as the start, stop and tally buttons.
- A repeater status board that shows the club repeater's ID schedule, linked status, and upcoming nets.
- A field-day band and score display, fed from the logging network.
- A shack sensor panel for temperature, humidity and supply voltage, using I2C sensors on the free pins.

The CYD also shows up in Wi-Fi security tools such as ESP32 Marauder and Bruce. Those are not ham projects, so steer new members to the radio dashboards above.

HamClock and the CYD

The original HamClock does not run on a CYD; it needs a Linux computer such as a Raspberry Pi. What runs on the CYD is the CYD Ham Dashboard, an independent project inspired by HamClock's look and purpose.

A short history

Elwood Downey, WB0OEW, introduced HamClock in an October 2017 QST article. It became one of the most familiar screens in ham shacks: space weather, propagation, DX spots and satellite passes on one kiosk display. HamClock was split into an open-source client and a closed backend server that supplied its data, which is why its future came into question in 2026.

Date	Event
Aug 2026	CYD Ham Dashboard by HenrysCat gains wide notice as a HamClock-inspired firmware for the CYD
June 2026	Original Clear Sky Institute backend stops serving data; HamClocks must point to a community backend
Feb 2026	hamclock.com (closed source) and the Open HamClock Backend, OHB (open source, led by Brian Wilkins KO4AQF), offer replacement servers; Inovato and N0LSR Design endorse OHB for their appliances
Jan 29, 2026	WB0OEW becomes a Silent Key; the HamClock site announces version 4.22 is the last release
Oct 2017	HamClock introduced in QST

Sources: ARNewsline, Amateur Radio Daily, Open HamClock on GitHub, Amateur Radio Daily on OHB, hamclock.com.

If you already run HamClock on a Pi

Your HamClock needs a new backend server. Point it to either hamclock.com or the Open HamClock Backend; both are free. The Open HamClock organization on GitHub also maintains the client and pre-built Raspberry Pi images.

What the CYD Ham Dashboard gives you

The CYD Ham Dashboard is open-source C++ firmware (GPL-3.0) that runs entirely on the ESP32. It needs no Pi, no SD card, and no HamClock backend, because it pulls data directly from public sources.

Page	Shows	Data source
Clock	UTC and local time, Wi-Fi status, IP address	NTP time servers
HF propagation	SFI, sunspot number, A and K index, band ratings 80–10 m	N0NBH HamQSL
VHF conditions	Aurora, Sporadic-E on 6/4/2 m, flare alerts	N0NBH HamQSL
Greyline map	Day/night terminator, sunrise and sunset, your QTH	Calculated on the board
PSKReporter map	Where your signal is being heard	PSKReporter
DX cluster	Live spots via telnet or web feed	Your chosen DX cluster
POTA spots	Activators, optionally filtered by distance	POTA

Sources: hamradio.my review (9M2PJU), Open Source For You.

It supports the 2.8-inch CYD with either display driver and a 4-inch 480×320 variant. Tap the left or right side of the screen to change pages. It draws under 1 W, versus roughly 5–15 W for a Pi and monitor.

What it does not do: VOACAP propagation maps, satellite tracking, or the dense multi-pane layout of the full HamClock. On a 2.8-inch screen you see one page at a time.

Step by step: flashing the CYD Ham Dashboard

The whole job takes about ten minutes and needs only a computer with Chrome or Edge, a USB data cable, your callsign, and your Maidenhead grid square.

1. **Open the flasher.** In Chrome or Edge on a desktop computer, go to henryscat.github.io/esp32-cyd-ham-dashboard. The HenrysCat flasher index lists the author's other CYD firmware too.
2. **Pick your board.** Choose ILI9341 for the Elegoo 2.8-inch board and the original ESP32-2432S028R. If the image comes up mirrored, keep this build and turn on Mirror display in the settings page (step 8). Choose ST7789 only if the ILI9341 build leaves the screen blank or garbled; newer E32R28T-1 boards use an ST7789 panel. Choose ST7796S for the 4-inch 480×320 board.
3. **Connect and install.** Plug the CYD into the computer, click the install button, and pick the new serial port (often labeled CH340 or CP2102).
4. **Erase on first install.** When asked, tick Erase device for a first install. On later upgrades, leave it unticked to keep your callsign, locator and Wi-Fi settings.
5. **Join the setup hotspot.** After the board reboots, it starts its own Wi-Fi setup network. Connect a phone to it; a setup page should open on its own (if not, browse to 192.168.4.1).
6. **Enter your details.** Fill in your home Wi-Fi name and password, callsign, 6-character grid square, and time zone, then save.
7. **Let it populate.** The board joins your network, sets its clock, and fills in each page over the next minute or two. Tap the right or left edge of the screen to move between pages.
8. **Fine-tune from a browser.** Any computer on your network can open the settings page at the address shown on the clock page (the 9M2PJU review gives cyd-ham.local; the author found the board by its IP address instead, as noted

below). There you can change brightness, the DX cluster server, the POTA distance filter, and color or rotation fixes.

Notes from testing the Elegoo board:

- **The image came up mirrored.** The fix is in the dashboard's settings page: open it and turn on Mirror display.
- **Finding the IP address.** If the board's name does not resolve in your browser (common on Windows), use a free network scanner such as Advanced IP Scanner to find the board's IP address, then type that address into the browser. Your router's list of connected devices works too.

Per the 9M2PJU review, holding the BOOT button for about five seconds while the dashboard runs resets it to factory settings. The author has not tested this on the Elegoo board.

Flasher details are from the official flasher page; setup and reset details are from the 9M2PJU review and Open Source For You. Check the project's GitHub README for the current hotspot name and password before a club build night.

Enclosures, power and RF in the shack

Put the board in a case, feed it clean 5 V, and treat it like any other digital gadget near your receiver.

Enclosures

A bare board on a desk full of coax and tools invites a short. The Elegoo kit includes an acrylic case with mounting screws, which is enough for most desks. For an angled stand or a sturdier field case, free 3D-printable designs are plentiful; two good starting points are an angled desktop enclosure with a stylus holder and a compact snap-fit case for portable use. If no one in the club owns a printer, a small project box with a window cut for the screen works fine. Tilt the display toward your eyes, since the TN panel dims when viewed off-axis.

Power

The CYD runs on 5 V USB and draws roughly 120–180 mA, or under 1 W (9M2PJU review). Options include a USB port on a computer or hub, a phone charger, a power bank for field use, or a 12 V-to-5 V converter off the station supply. Use a supply that delivers a steady 5 V; marginal supplies cause random resets when Wi-Fi transmits.

RF considerations

- **Receive noise.** Cheap USB chargers and buck converters can radiate HF hash. If you hear new noise, try a linear supply or a better charger, and add clip-on ferrite chokes on the USB cable near both ends.
- **The board's own emissions.** The ESP32's 2.4 GHz Wi-Fi sits close to the 13 cm amateur band; keep the CYD away from 2.4 GHz gear and its antennas. The digital circuitry can also produce weak spurs on VHF/UHF; moving it a meter from the rig usually solves that.
- **Your transmitter.** Strong RF in the shack can reset the board or upset the touchscreen. Good station grounding, chokes on the USB lead, and keeping the CYD away from the antenna feed point all help.
- **Wi-Fi coverage.** The board needs your shack Wi-Fi. If the signal is weak, move it closer to the access point or add a small extender; dropped connections show up as stale data.

Troubleshooting

Most CYD problems come down to the cable, the board revision, or Wi-Fi. Work down this table before assuming the board is bad.

Symptom	Likely cause	Fix
Computer does not see the board	Charge-only cable or missing driver	Swap to a known data cable; install the CH340 driver
Flasher has no Install button or port list	Unsupported browser	Use Chrome or Edge on a desktop computer
Flashing stalls at "Connecting"	Board not entering download mode	Hold BOOT while clicking Install, release when it starts
Screen blank or white after flashing	Wrong display driver build	Reflash with the other driver option (ILI9341 vs ST7789)
Colors wrong or image mirrored	Panel variant differences	Turn on Mirror display in the settings page (needed on the author's Elegoo board); color and rotation fixes are there too
Touch hits the wrong spot	Resistive touch needs calibration or axis flip	Run the firmware's calibration or flip touch axes in settings
Random reboots	Weak power supply or RF pickup	Better 5 V supply; ferrite chokes; move away from antennas
Pages show old or no data	Wi-Fi drop or data source down	Check signal strength on the clock page; try a manual refresh
Not sure what board you have	Unlabeled clone	Flash the HenrysCat CYD Diagnostic Tool to identify and test it

When all else fails, the witnessmenow repository and its Discord community are friendly to beginners.

Sources and further reading

All pages below were checked on 2026-11-02. Vendor prices change quickly; confirm them before printing.

Hardware

- atomic14: Cheap Yellow Display family guide
- atomic14: ESP32-2432S028R board page
- witnessmenow/ESP32-Cheap-Yellow-Display (community hub)
- Random Nerd Tutorials: Getting started with the CYD

ESP32-32E board (E32R28T)

- LCDWIKI: 2.8inch ESP32-32E (E32R28T) Specification V1.0 (PDF); the E32R28T-1 revision page covers the newer ST7789 version
- Home Assistant community: E32R28T display setup thread

CYD Ham Dashboard and other firmware

- CYD Ham Dashboard on GitHub (HenrysCat)
- CYD Ham Dashboard web flasher
- HenrysCat flasher index (diagnostic tool, weather dashboard)
- 9M2PJU review of the CYD Ham Dashboard (hamradio.my)
- Open Source For You: ESP32 CYD runs ham radio dashboard
- Matt Housley cyd-dashboard (Trendshift listing)
- 9M2PJU ESP32 DX Cluster Client

HamClock history and backends

- ARNewsline: Silent Key Elwood Downey, WB0OEW
- Amateur Radio Daily: HamClock project to sunset June 2026
- Amateur Radio Daily: Inovato, N0LSR Design endorse OHB
- Open HamClock on GitHub
- hamclock.com community backend

Enclosures

- Printables: CYD enclosure with pen holder
- Printables: compact CYD case